[Total: 6]

2 A student uses a motion-sensor connected to a laptop to investigate the motion of a hollow ball of mass 1.2×10^{-2} kg falling through air.

The ball is dropped from rest. It reaches terminal velocity before it reaches the ground.

The upthrust on the ball is negligible.
Fig. 2.1 shows the variation with time t of the velocity v of the ball as it falls towards the ground

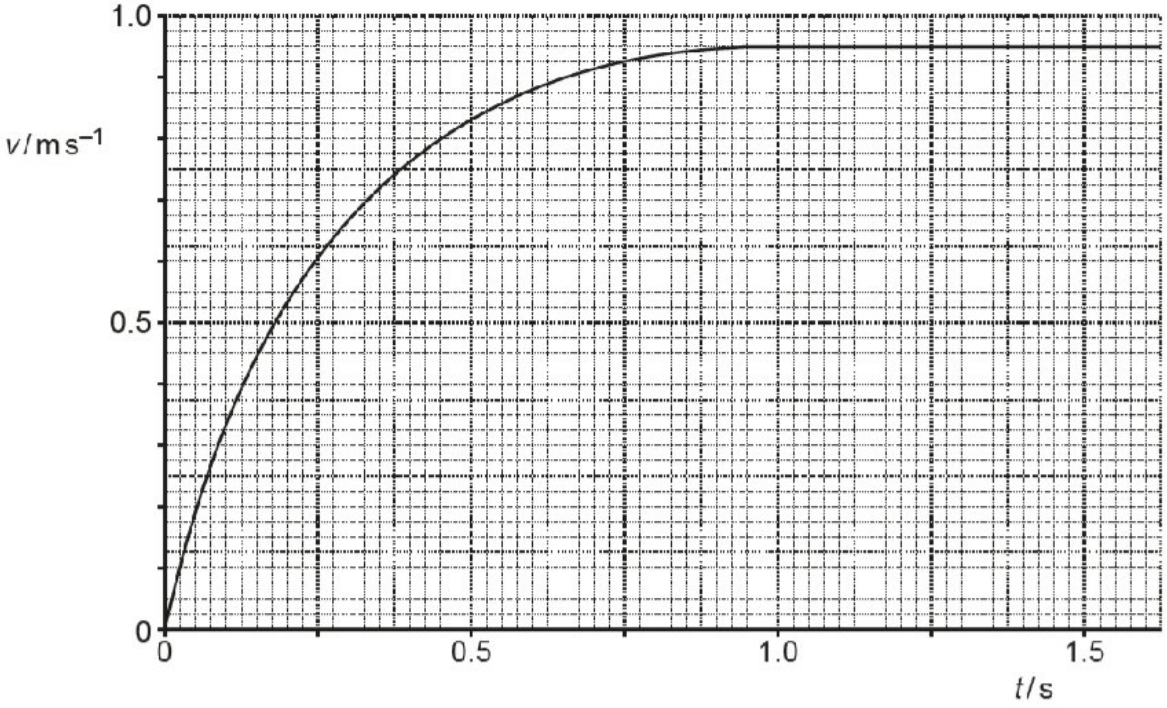


Fig. 2.1

(a) Using Fig. 2.1, state and explain how the acceleration of the falling object changes over the first 1.5 s.

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..... [2]

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(b) Use the graph to determine the acceleration of the object at time t = 0.25 s after it was released.

acceleration = ... m s^{-2} [2]

(c) Use your answer in **(b)** to determine the drag force acting on the ball at time t = 0.25 s

drag force = ... N [2]

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(d) The student now adds a small amount of sand inside the hollow ball. As before, the ball is dropped from rest and it also reaches terminal velocity before it reaches the ground.

(i) State and explain how the forces acting on the sand-filled ball at v = 0.50 m s^{-1} compares with the forces acting on the hollow ball at this speed.

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(ii) State and explain how the terminal velocity of the sand-filled ball compares with the terminal velocity of the hollow ball.

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..... [2]

[Total :10]